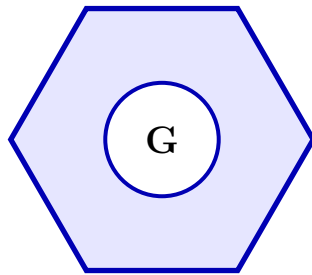


Gono Protocol

*A Modular Blockchain Infrastructure for
Verifiable Media and Autonomous Commerce*

Technical Whitepaper

Version 0.1



Gono Protocol Foundation

February 5, 2026

*This document serves as a technical specification for the Gono Protocol,
designed to meet the enablement requirements for intellectual property protection.*

Abstract

The present disclosure relates generally to distributed ledger systems for digital media provenance, and more specifically to a modular blockchain protocol that provides immutable indexing, verification, and micropayment capabilities for content-addressable assets.

The GONO Protocol is an original, modular blockchain infrastructure built on Substrate as a Polkadot Parachain. The protocol is designed as a universal, content-addressable rail that provides an immutable foundation for verifiable media, digital assets, and autonomous artificial intelligence (AI) commerce. By utilizing a pluggable module architecture, the protocol enables developers to construct high-security journalism platforms, lightweight asset trackers, or automated AI data marketplaces using a unified core infrastructure.

The system comprises four distinct architectural layers: (1) a shared security layer leveraging the Polkadot Relay Chain for institutional-grade protection against censorship and majority attacks; (2) a core execution rail implementing ERC-7053 indexing logic for media receipt management; (3) modular service pallets providing optional functionality including permanent storage via Arweave integration, community verification through SANUB reputation algorithms, privacy preservation using zero-knowledge proofs, and machine-to-machine micropayments via the x402 protocol; and (4) an application layer supporting user-facing portals through decentralized APIs.

This whitepaper provides a comprehensive technical specification of the protocol architecture, operational workflows, module interfaces, economic mechanisms, and governance structures.

Keywords: Blockchain, Media Provenance, Digital Asset Verification, Micropayments, Zero-Knowledge Proofs, Content Addressability, Polkadot, Substrate, ERC-7053, x402 Protocol

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1 Field of the Invention

The present disclosure relates generally to distributed ledger technology and blockchain systems, and more specifically to modular blockchain protocols designed for digital media provenance tracking, content verification, and autonomous machine-to-machine commerce.

2 Background of the Invention

2.1 Technical Context

The proliferation of digital media and the rise of artificial intelligence have created unprecedented challenges in establishing the authenticity, provenance, and ownership of digital content. Conventional systems for media verification suffer from centralization vulnerabilities, lack of permanence, and insufficient mechanisms for automated commerce between AI agents and human participants.

2.2 Limitations of Existing Approaches

Existing approaches to digital media verification and blockchain-based provenance tracking exhibit several deficiencies:

- (a) **Centralization Risks:** Traditional verification systems rely on centralized authorities, creating single points of failure susceptible to censorship, manipulation, and state-level interference.
- (b) **Lack of Permanence:** Conventional storage solutions do not guarantee long-term availability of content, leaving verified media vulnerable to deletion or modification.
- (c) **Privacy Limitations:** Existing blockchain systems typically require full disclosure of participant identities, creating risks for high-stakes applications such as investigative journalism.
- (d) **Inflexible Architecture:** Monolithic blockchain designs force applications to adopt full protocol functionality regardless of specific use-case requirements, resulting in unnecessary overhead.
- (e) **Inadequate Micropayment Support:** Current blockchain systems lack native support for efficient machine-to-machine micropayments required for AI agent commerce.

2.3 Need for Improvement

There exists a need for a modular blockchain infrastructure that provides developers with the flexibility to selectively implement verification, storage, privacy, and payment functionalities while maintaining a unified, secure foundation for media provenance and digital commerce.

3 Summary of the Invention

The present disclosure provides a modular blockchain protocol, referred to herein as the GONO Protocol, designed to address the limitations described above. The protocol comprises a layered architecture that enables selective adoption of functionality while maintaining institutional-grade security through integration with the Polkadot ecosystem.

3.1 Primary Objectives

An objective of the present disclosure is to provide a content-addressable blockchain rail for immutable media provenance tracking.

A further objective is to provide modular service pallets that may be selectively enabled based on application requirements.

A still further objective is to enable zero-knowledge proof-based anonymity for high-stakes use cases while maintaining verifiable authenticity.

Yet another objective is to provide native support for machine-to-machine micropayments through implementation of the x402 protocol standard.

An additional objective is to establish a robust governance framework that balances economic stakeholder interests with meritocratic expertise-based decision-making for content integrity.

3.2 Technical Solution Overview

In accordance with embodiments of the present disclosure, a blockchain protocol is provided comprising:

- A shared security layer utilizing the Polkadot Relay Chain;
- A core execution rail implementing ERC-7053 indexing for media receipts;
- A plurality of optional modular pallets including storage, verification, privacy, and micropayment modules;
- A comprehensive governance framework with token-weighted, reputation-weighted, and AI-native decision-making tracks;
- An application layer providing decentralized API access to protocol functionality.

4 Brief Description of the Drawings

FIG. 1 is a block diagram illustrating the high-level modular architecture of the GONO Protocol according to an embodiment of the present disclosure.

FIG. 2 is a block diagram illustrating the modular pallet architecture according to an embodiment of the present disclosure.

FIG. 3 is a flowchart describing the integrated “Gono Flow” pipeline for content provenance according to an embodiment of the present disclosure.

FIG. 4 is a block diagram illustrating the multi-provider storage architecture with dynamic endowment funding according to an embodiment of the present disclosure.

FIG. 5 is a sequence diagram illustrating the x402 micropayment flow according to an embodiment of the present disclosure.

FIG. 6 is a block diagram illustrating the token economic model according to an embodiment of the present disclosure.

FIG. 7 is a block diagram illustrating the three-tier Gono Sovereign Governance architecture according to an embodiment of the present disclosure.

FIG. 8 is a diagram illustrating the SANUB reputation algorithm flow, showing the relationship between user roles and calculated credibility scores according to an embodiment of the present disclosure.

5 Detailed Description of the Invention

The following detailed description provides a comprehensive explanation of the GONO Protocol architecture, components, and operational workflows. Reference numerals are used throughout to identify components illustrated in the accompanying figures.

5.1 High-Level Modular Architecture

Referring now to FIG. 1, the GONO Protocol (100) is structured into four distinct architectural layers, enabling maximum flexibility and performance optimization.

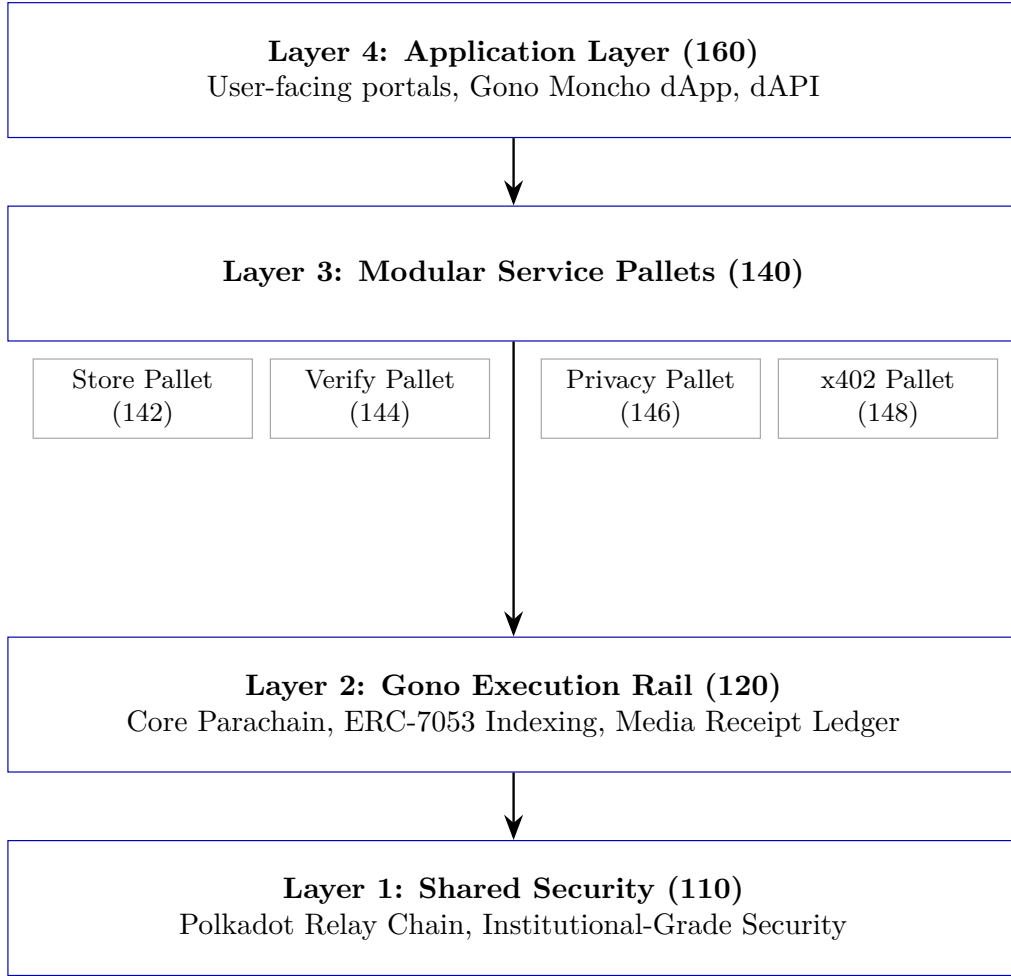


Figure 1: High-Level Modular Architecture of the Gono Protocol (100)

5.1.1 Layer 1: Shared Security Layer (110)

The Shared Security Layer (110) leverages the Polkadot Relay Chain to provide institutional-grade security from initial deployment. This configuration ensures that the GONO Protocol (100) exhibits resistance to state-level censorship and 51% attacks. The Relay Chain provides:

- Shared validator set for consensus security;
- Cross-chain messaging capabilities via Cross-Consensus Messaging (XCM);
- Pooled security resources across the Polkadot ecosystem.

5.1.2 Layer 2: Gono Execution Rail (120)

The Gono Execution Rail (120) comprises a specialized Substrate-based blockchain operating as a Polkadot Parachain. The Execution Rail (120) handles mandatory ERC-7053 indexing logic and functions as the global “Media Receipt” ledger.

The Execution Rail (120) implements:

- ERC-7053 compliant metadata indexing;
- Content-addressable hash storage;
- Version control for media assets;
- Timestamp verification and ordering.

5.1.3 Layer 3: Modular Service Pallets (140)

The Modular Service Pallets (140) provide optional extensions that developers may selectively enable based on application requirements. The pallet architecture enables a “plug-in” approach where functionality is composed rather than inherited.

Store Pallet (142) The Store Pallet (142) provides a multi-provider bridge for permanent content hosting across Arweave, Filecoin, and StorJ networks. The Store Pallet (142) implements a “pay-once, store-forever” economic model with Storage SLA guarantees, wherein users pay a Storage Endowment to fund permanent file hosting. For high-value assets, a “Top-up” mechanism enables community contributions to extend storage life beyond initial endowment projections.

Verify Pallet (144) The Verify Pallet (144) provides an agnostic interface for community attestations. The Verify Pallet (144) implements the SANUB (Sharing and Analyzing News Using Blockchain) framework [1] for content credibility calculation, with optional integration of AI oracles for hybrid verification approaches. For high-impact content, the Verify Pallet (144) may interface with a Journalistic Integrity Council for binding verdicts.

Privacy Pallet (146) The Privacy Pallet (146) implements zero-knowledge Succinct Non-interactive Arguments of Knowledge (zk-SNARKs) to enable proof of personhood or credential verification while maintaining wallet anonymity. The Privacy Pallet (146) may integrate with decentralized identity providers such as BrightID for humanity verification without exposing user identity on the public ledger.

x402 Micropayment Pallet (148) The x402 Micropayment Pallet (148) implements the x402 Open Standard, which revives HTTP-402 (Payment Required) for machine-to-machine (M2M) payments. The x402 Micropayment Pallet (148) enables AI agents to pay for data or services on a per-request basis without account creation.

5.1.4 Layer 4: Application Layer (160)

The Application Layer (160) comprises user-facing portals and applications that interface with the protocol via a decentralized API (dAPI). Exemplary applications include the Gono Moncho News dApp, which provides a decentralized journalism platform leveraging the protocol’s provenance and verification capabilities.

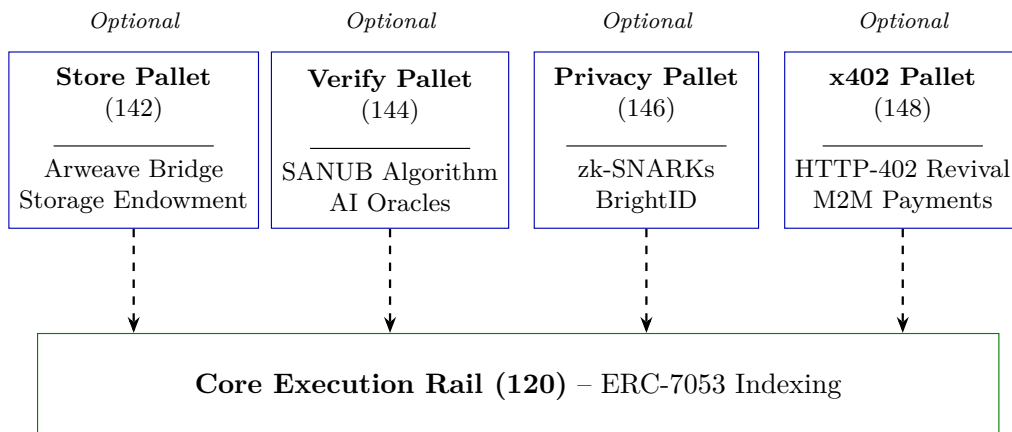


Figure 2: Modular Pallet Architecture showing optional service extensions

5.2 The Integrated Gono Flow

Referring now to FIG. 3, the GONO Protocol (100) implements a standardized pipeline for content provenance, wherein high-resource modules (Store and Verify) are optional. The underlying x402 Micropayment capability operates as a protocol-wide service facilitating commerce at any stage but does not constitute a standalone step in the provenance chain.

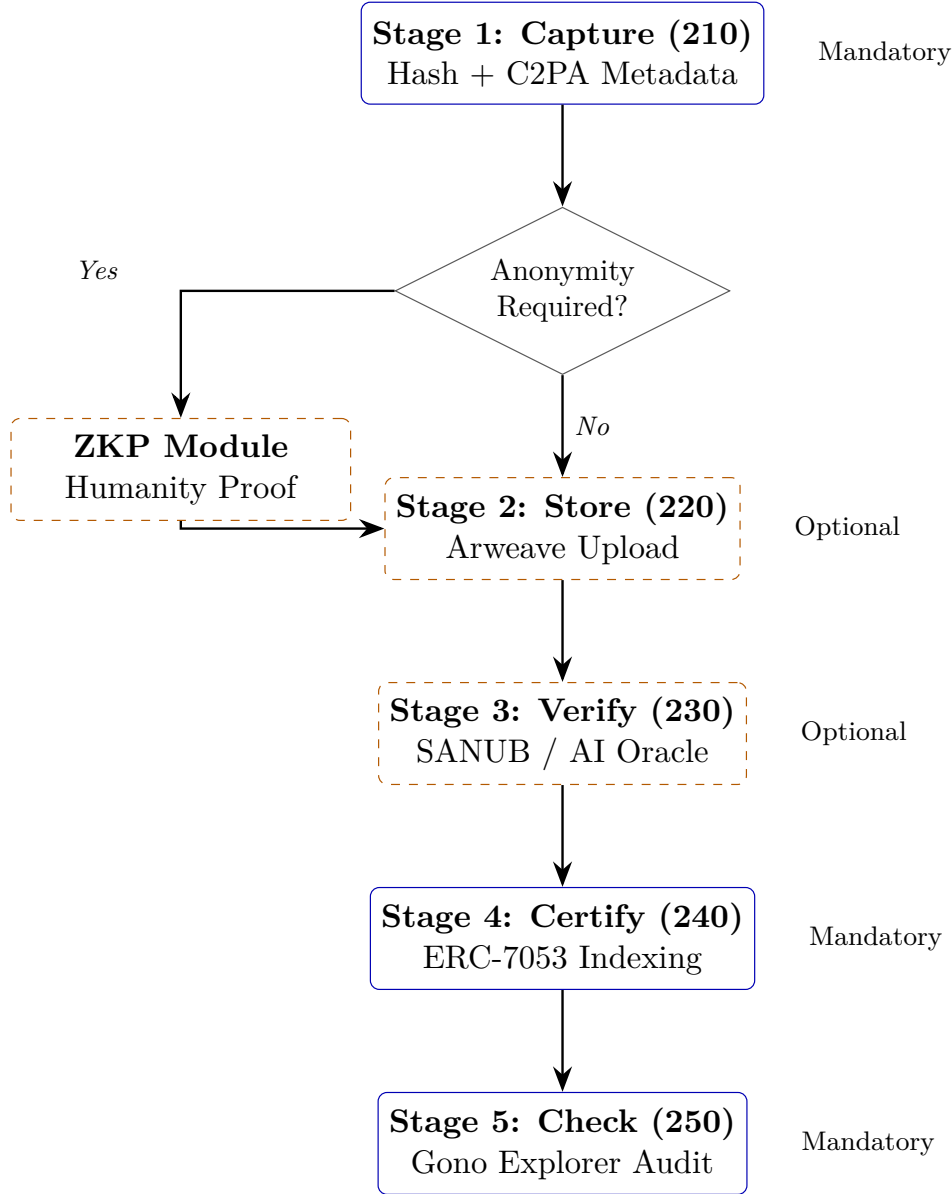


Figure 3: Integrated Gono Flow Pipeline for Content Provenance

5.2.1 Stage 1: Capture (210)

The Capture stage (210) is mandatory for all content entering the GONO Protocol (100). During the Capture stage (210), a Capture SDK generates a cryptographic hash of the digital asset and embeds Coalition for Content Provenance and Authenticity (C2PA) metadata.

In embodiments requiring anonymity (e.g., high-stakes journalism), the ZKP Module generates a proof of “humanity” without exposing the creator’s identity on the public ledger. The ZKP Module may interface with the Privacy Pallet (146) to verify credentials from services such as BrightID.

The Capture stage (210) produces:

- A cryptographic hash (e.g., SHA-256) of the original content;
- C2PA-compliant metadata including timestamp, device information, and creator attestations;
- Optionally, a zero-knowledge proof of personhood.

5.2.2 Stage 2: Store (220)

The Store stage (220) is optional and provides permanent content hosting for assets requiring long-term availability. When the Store stage (220) is invoked, the Store Pallet (142) coordinates storage across multiple decentralized providers.

Multi-Provider Storage Redundancy To protect against the failure of any single storage network’s economic model, the Store Pallet (142) implements concurrent storage across multiple decentralized providers:

- **Primary Storage:** Arweave (permanent, endowment-based);
- **Secondary Storage:** Filecoin (incentivized, contract-based);
- **Tertiary Storage:** StorJ (distributed, erasure-coded).

Cross-protocol replication enables a “Storage SLA” guarantee, wherein content remains accessible as long as at least one provider network remains operational. The Store Pallet (142) monitors provider availability and automatically rebalances storage allocation when network conditions change.

Storage Endowment Model The base storage model requires a one-time “Storage Endowment” in GONO tokens to fund the file’s hosting in perpetuity. This economic model ensures that stored content cannot be censored or deleted due to non-payment.

Dynamic Endowment Adjustments For high-value assets requiring extended guarantees, the Store Pallet (142) implements a “Top-up” mechanism:

- **Periodic Assessment:** Storage costs are reassessed at configurable intervals based on network economics and provider fee structures;
- **Community Contributions:** Users or communities may contribute additional GONO tokens to extend the storage life of critical historical records;
- **Endowment Pool:** Accumulated top-up contributions are managed through a dedicated endowment pool with transparent on-chain accounting;
- **Priority Tiers:** Assets may be designated as “Heritage” or “Critical Infrastructure” to receive priority consideration for community-funded storage extensions.

This dynamic model moves beyond static, front-loaded payments to ensure that culturally significant or high-impact content maintains permanent availability even as economic conditions evolve.

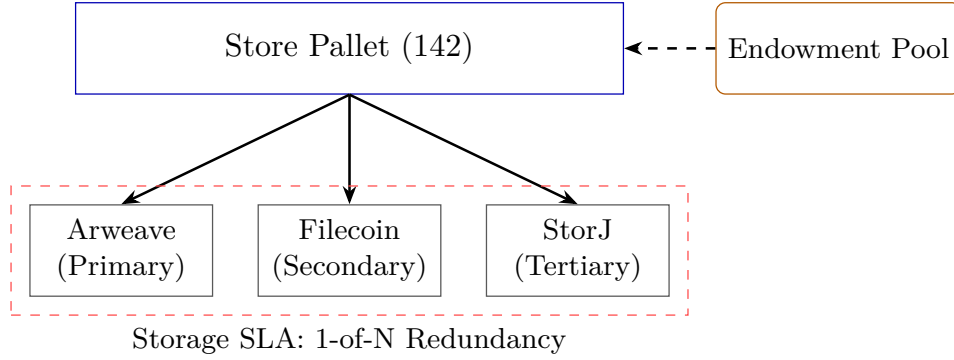


Figure 4: Multi-Provider Storage Architecture with Dynamic Endowment Funding

5.2.3 Stage 3: Verify (230)

The Verify stage (230) is optional and provides community or automated assessment of content authenticity. When the Verify stage (230) is invoked, the asset is submitted for assessment by community verifiers, automated AI Oracles, or both.

Verification results are processed through the SANUB reputation engine, which calculates credibility scores based on verifier history and consensus. For high-impact stories, the Journalistic Integrity Council may provide a final binding verdict.

5.2.4 Stage 4: Certify (240)

The Certify stage (240) is mandatory and creates the permanent on-chain record. During the Certify stage (240), the final state—including storage links, verification results, and metadata—is indexed via ERC-7053, creating a permanent “media receipt.”

The Certify stage (240) functions as a decentralized version control system analogous to Git, allowing users to append new evidence, updates, or certifications to an asset’s history without overwriting or deleting previous records. Each certification creates a new node in the asset’s provenance chain.

5.2.5 Stage 5: Check (250)

The Check stage (250) is mandatory and enables public auditing and transparency. Using the Gono Explorer, users, partners, or automated systems may cross-reference a file’s cryptographic hash against the on-chain registry to confirm authenticity.

The Gono Explorer provides a visual timeline of the asset’s complete, versioned journey, enabling verification that content is untampered and trustworthy.

5.3 Detailed Module Specification

Table 1 provides a comprehensive overview of the protocol’s technical modules and their associated standards.

Table 1: Technical Module Specification

Module	Technical Standard	Primary Purpose
Provenance Rail	ERC-7053	Immutable indexing of media receipts and metadata history. Provides content-addressable storage references and version control functionality.
Micropayments	x402 Protocol	Enables AI agents and humans to pay for resources per request without account creation. Revives HTTP-402 for machine-to-machine commerce.
Permanent Store	Arweave + Filecoin + StorJ	Multi-provider storage with SLA guarantees and dynamic Top-up mechanism for critical content.
Privacy Engine	zk-SNARKs	Cryptographic anonymity for high-stakes journalism or secure data sharing. Proves personhood without revealing identity.
Trust Logic	SANUB Algorithm	Blockchain-based framework for content credibility calculation through community verification, as defined by Balouchestani et al. [1].

5.4 x402 Micropayment Protocol Implementation

Referring now to FIG. 5, the x402 Micropayment Pallet (148) implements the x402 Open Standard for machine-to-machine payments.

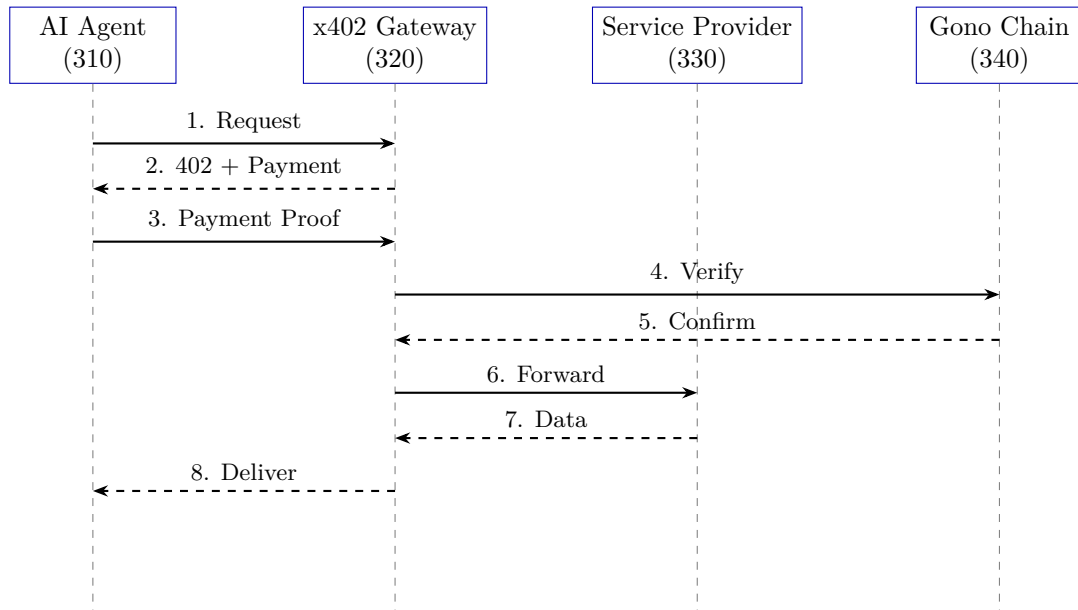


Figure 5: x402 Micropayment Flow Sequence Diagram

The x402 micropayment flow operates as follows:

1. The AI Agent (310) requests a resource from the x402 Gateway (320).
2. The x402 Gateway (320) returns an HTTP 402 response with payment details including price, acceptable payment methods, and payment address.
3. The AI Agent (310) submits a payment proof to the x402 Gateway (320).
4. The x402 Gateway (320) verifies the payment on the Gono Chain (340).
5. The Gono Chain (340) confirms the payment validity.
6. The x402 Gateway (320) forwards the original request to the Service Provider (330).
7. The Service Provider (330) returns the requested resource data.
8. The x402 Gateway (320) delivers the resource to the AI Agent (310).

While x402 typically settles in stablecoins (such as USDC), **GONO** may be used as the protocol-level fee (“provenance gas”) for every micropayment receipt generated on-chain.

6 Economic and Governance Logic

The GONO Protocol (100) utilizes the **GONO** token as its native economic engine. Referring to FIG. 6, the token serves multiple functions within the protocol ecosystem.

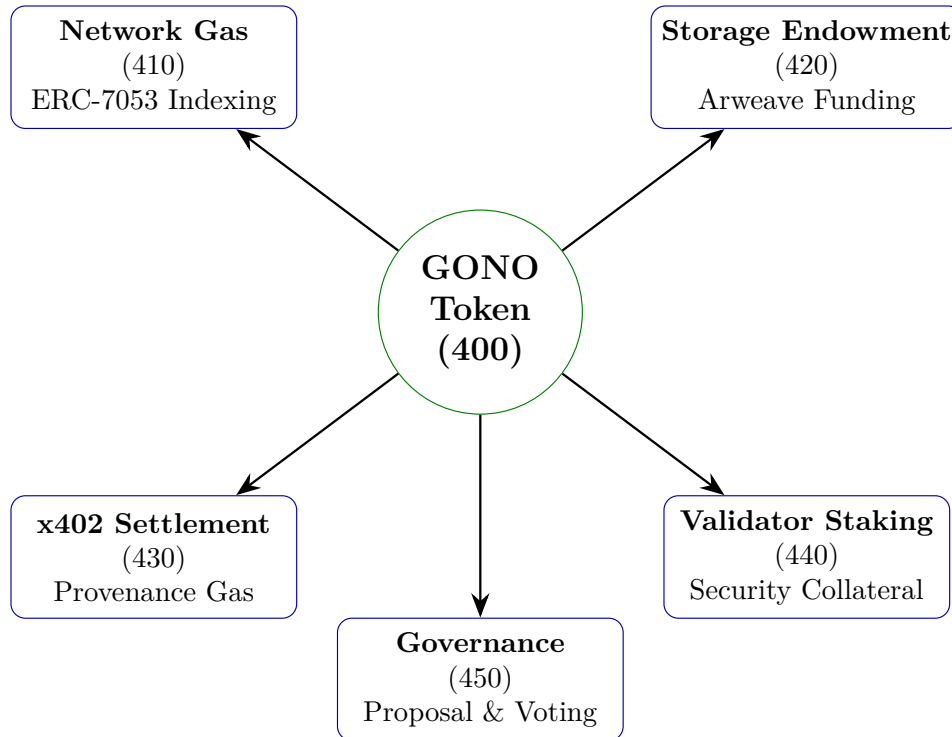


Figure 6: GONO Token Economic Model showing utility functions

6.1 Token Utility Functions

6.1.1 Network Gas (410)

All ERC-7053 on-chain indexing operations require payment in **GONO** tokens. This includes the mandatory Capture (210), Certify (240), and Check (250) stages of the Gono Flow.

6.1.2 Storage Endowment (420)

Users invoking the Store stage (220) pay a one-time upfront fee in **GONO** tokens to secure permanent storage space on Arweave. The Storage Endowment is calculated based on file size

and current network conditions.

6.1.3 x402 Settlement Support (430)

While x402 micropayments may settle in stablecoins (such as USDC), GONO serves as the protocol-level fee for generating micropayment receipts. Each x402 transaction incurs a small GONO fee for provenance tracking.

6.1.4 Validator Staking (440)

Nodes securing the modular pallets are required to stake GONO tokens as collateral. Validators are subject to slashing conditions for:

- Poor service availability or performance;
- Fraudulent attestations in the Verify Pallet (144);
- Protocol violations or censorship attempts.

6.1.5 Governance (450)

The GONO Protocol implements a hybrid governance framework termed “Gono Sovereign Governance,” which blends token-weighted economic governance, reputation-weighted integrity governance, and AI-native autonomous governance. This three-tier model ensures that protocol evolution is guided by stakeholders with aligned incentives and demonstrated expertise.

Tier 1: Core Economic Governance (Token-Weighted) Core economic governance handles fundamental protocol parameters including GONO token inflation, slashing conditions for validators, and treasury allocations. This tier utilizes Conviction Voting, inspired by Polkadot OpenGov, to reward long-term stakers.

Table 2: Core Economic Governance Parameters

Parameter	Threshold	Rationale
Proposal Threshold	0.5% of total GONO supply	Balances accessibility with spam protection
Quorum	10% of total GONO supply	Ensures broad consensus for fundamental changes
Approval Threshold	>50% (Simple Majority)	Standard democratic benchmark
Timelock	7 Days	Mandatory cooling-off period for economic security

Staked GONO tokens confer rights to:

- Propose technical upgrades and protocol modifications;
- Vote on new module parameters and configurations;
- Determine fee schedules and economic parameters;
- Establish validator Service Level Agreements (SLAs).

Tier 2: Technical and Integrity Governance (Reputation-Weighted) Technical and Integrity Governance governs the Verify Pallet (144), SANUB algorithm parameters, and the Journalistic Integrity Council. Voting power in this tier is derived from a user’s Analyst Credit (C_a) rather than token holdings, ensuring that “whales” cannot manipulate the reputation scoring of news items.

The voting power in the Integrity Track is calculated using a reputation-weighted formula:

$$V_{integrity} = C_a \times e^{-\lambda(t-t_{last})} \quad (1)$$

Where C_a is the analyst’s credit score, λ is the decay factor, and $(t - t_{last})$ is the time since the analyst’s last active verification. This meritocratic approach ensures that governance is led by those who contribute most to the protocol’s accuracy.

Journalistic Integrity Council The Journalistic Integrity Council is a specialized Technical Fellowship composed of analysts who have achieved a minimum Analyst Credit threshold ($C_a \geq 0.8$). The Council has authority to:

- Issue binding verdicts on high-impact contested content;
- Propose modifications to SANUB algorithm parameters;
- Whitelist emergency security patches for the Verify Pallet (144);
- Define content moderation policies and ethical guidelines.

Council members are selected through a merit-based system where the top 50 analysts by C_a score are automatically granted Council membership, with a minimum of 100 verified analyses required for eligibility.

Tier 3: Autonomous Commerce Governance (AI-Native) Given the x402 implementation, the GONO Protocol requires a governance track for machine-to-machine commerce. This tier manages the x402 Facilitator whitelist, service discovery parameters, and storage top-up priorities.

Decisions in this track may be partially automated, using AI agents to vote on Service Level Agreement (SLA) parameters based on network congestion and provider availability. Human oversight is maintained through veto power held by Tier 1 token holders.

Dispute Resolution: Kleros Integration For subjective disputes arising when the SANUB algorithm reaches a contested or low-confidence verdict, the GONO Protocol integrates with Kleros decentralized arbitration:

1. **Escalation:** If a Verify request (Stage 230) results in a contested verdict, any party may stake GONO to escalate the dispute to a specialized Kleros court.
2. **Juror Selection:** Jurors who have staked PNK tokens in the “Journalism” sub-court are randomly selected using cryptographic sortition.
3. **Evidence Review:** Jurors examine C2PA metadata, evidence submitted by Reporters and Analysts, and SANUB scoring history.
4. **Enforcement:** The final Kleros ruling is automatically executed on-chain via the Gono Execution Rail (120), updating the asset’s “media receipt” in Stage 4 (Certify).

This “Epistemic Engine” ensures that truth determination is governed through game-theoretic incentives rather than centralized authority.

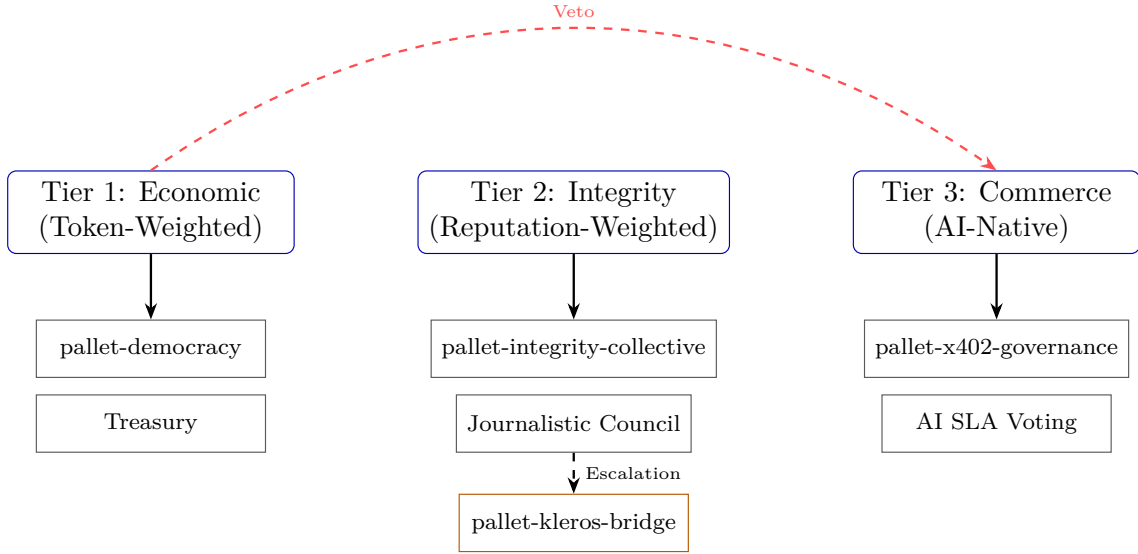


Figure 7: Gono Sovereign Governance: Three-Tier Architecture

6.2 Economic Flow Summary

Table 3 summarizes the economic flows within the protocol.

Table 3: Protocol Economic Flows

Action	Fee Type	Currency	Recipient
Content Indexing	Gas Fee	GONO	Network Validators
Permanent Storage	Endowment	GONO	Arweave + Protocol
Verification Request	Service Fee	GONO/Stablecoin	Verifiers
x402 Micropayment	Provenance Gas	GONO	Protocol Treasury
Governance Proposal	Deposit	GONO	Escrow (Refundable)

7 Alternative Embodiments

The present disclosure contemplates various alternative embodiments and modifications within its scope.

7.1 Layer 1 Alternatives

While the preferred embodiment utilizes Polkadot as the shared security layer, the GONO Protocol (100) may alternatively be implemented on:

- Cosmos ecosystem as an IBC-enabled chain;
- Ethereum as a Layer 2 rollup;
- A standalone proof-of-stake blockchain;
- A consortium or permissioned network for enterprise deployments.

7.2 Storage Alternatives

The Store Pallet (142) may interface with alternative permanent storage solutions including:

- InterPlanetary File System (IPFS) with Filecoin incentivization;
- Sia decentralized storage network;

- StorJ distributed storage platform;
- Enterprise storage solutions for private deployments.

7.3 Privacy Alternatives

The Privacy Pallet (146) may implement alternative zero-knowledge proof systems including:

- zk-STARKs for quantum-resistant proofs;
- Bulletproofs for range proofs and confidential transactions;
- PLONK for efficient universal SNARKs;
- Groth16 for optimized pairing-based proofs.

7.4 Verification Alternatives

The Verify Pallet (144) may support alternative verification mechanisms including:

- Decentralized oracle networks such as Chainlink or Band Protocol;
- Machine learning-based automated fact-checking;
- Federated verification across multiple news organizations;
- Prediction market-based truth discovery.

7.5 Implementation Variations

The modular pallets described herein may be implemented as:

- On-chain smart contracts;
- Off-chain services with on-chain anchoring;
- Hardware security modules (HSMs) for sensitive operations;
- Trusted execution environments (TEEs) such as Intel SGX;
- Cloud-based microservices with cryptographic attestation.

8 Technical Implementation Details

8.1 ERC-7053 Implementation

The ERC-7053 standard implemented by the Gono Execution Rail (120) provides the following capabilities:

- **Content Addressing:** Each media asset is identified by a cryptographic hash (e.g., SHA-256, Keccak-256) that serves as its unique content address.
- **Metadata Indexing:** C2PA-compliant metadata is indexed on-chain, enabling efficient lookup and verification.
- **Version Control:** Assets may have multiple versions, each linked through a directed acyclic graph (DAG) structure similar to Git commits.
- **Provenance Chain:** Each modification or attestation creates a new node in the asset's provenance chain, maintaining complete history.

8.2 SANUB Reputation Algorithm

The SANUB (Sharing and Analyzing News Using Blockchain) algorithm, as defined by Balouchestani et al. [1], provides a comprehensive framework for calculating content credibility through community participation. The GONO Protocol adapts this algorithm for the Verify Pallet (144) with the following mathematical foundations.

8.2.1 User Roles

SANUB defines three user roles within the verification ecosystem:

- **Reporters:** Users who publish content with supporting sources and evidence;
- **Analysts:** Users who evaluate content and provide analysis on validity;
- **Verifiers:** Users who score content based on their assessment (0 for fake, 1 for valid).

8.2.2 Public Belief Calculation

Given N_n users scoring a piece of content, where p_k is the score (0 or 1) from user k , the public belief B_n is calculated as:

$$B_n = \frac{\sum_{k=1}^{N_n} p_k}{N_n} \quad (2)$$

8.2.3 News Importance

The importance of content I_n is proportional to user participation, where N_T is the total registered users:

$$I_n = \frac{N_n}{N_T} \quad (3)$$

8.2.4 Belief Sigmoid Function

To normalize public belief within the range $[0, 1]$, a sigmoid function is applied:

$$S(B_n) = \frac{e^{(B_n - 0.75)}}{e^{(B_n - 0.75)} + 1} \quad (4)$$

8.2.5 Analyst Credit Calculation

Analyst credit C_a is computed based on accuracy of their analyses. Let a_p be the number of confirmed contents and a_n be the number of rejected contents by an analyst. The total positive credit is:

$$T_p = \sum_{i=1}^{a_p} S(B_{n_i}) + \sum_{j=1}^{a_n} S(1 - B_{n_j}) \quad (5)$$

Where $a_t = a_p + a_n$ is the total analyses. The analyst credit with punishment weighting is:

$$C_a = \frac{T_p}{T_p + (a_t - T_p)^{(2 + \frac{1}{a_t})}} \quad (6)$$

This formulation ensures that incorrect analyses are penalized more heavily than correct analyses are rewarded.

8.2.6 Reporter Credit Calculation

Reporter credit C_r incorporates analyst endorsements and content importance. For a reporter with n_{rt} published contents:

$$C_r = \sum_{i=1}^{n_{rt}} \left(\frac{\sum_{j=1}^{a_{p_i}} C_{a_j}}{\sum_{j=1}^{a_{p_i}} C_{a_j} + \left(\sum_{k=1}^{a_{n_i}} C_{a_k} \right)^{(2 + \frac{1}{a_{t_i}})}} \times I_{n_i} \right) \quad (7)$$

Where a_{p_i} and a_{n_i} are the analysts who approved and rejected the i -th content respectively, and a_{t_i} is the total analysts for that content.

8.2.7 Content Credibility Score

The final content credibility C_n combines reporter credit with analyst endorsements:

$$C_n = \frac{\sum_{i=1}^{a_{pn}} C_{a_i} \times C_r}{\sum_{i=1}^{a_{pn}} C_{a_i} + \sum_{k=1}^{a_{nn}} C_{a_k}} \quad (8)$$

Where a_{pn} is the number of supporting analysts and a_{nn} is the number of rejecting analysts.

8.2.8 Security Properties

The SANUB algorithm exhibits resistance to the following attack vectors:

- **Collusion Resistance:** Analysts cannot effectively collude with verifiers due to the large, anonymous verifier pool. Supporting fake content reduces analyst credit.
- **Sybil Resistance:** New identities have low initial credit, limiting their impact on calculations. Integration with the Privacy Pallet (146) for humanity verification further mitigates Sybil attacks.
- **Spam Prevention:** Publishing numerous low-importance but accurate contents does not significantly increase reporter credit due to the importance weighting factor I_n .

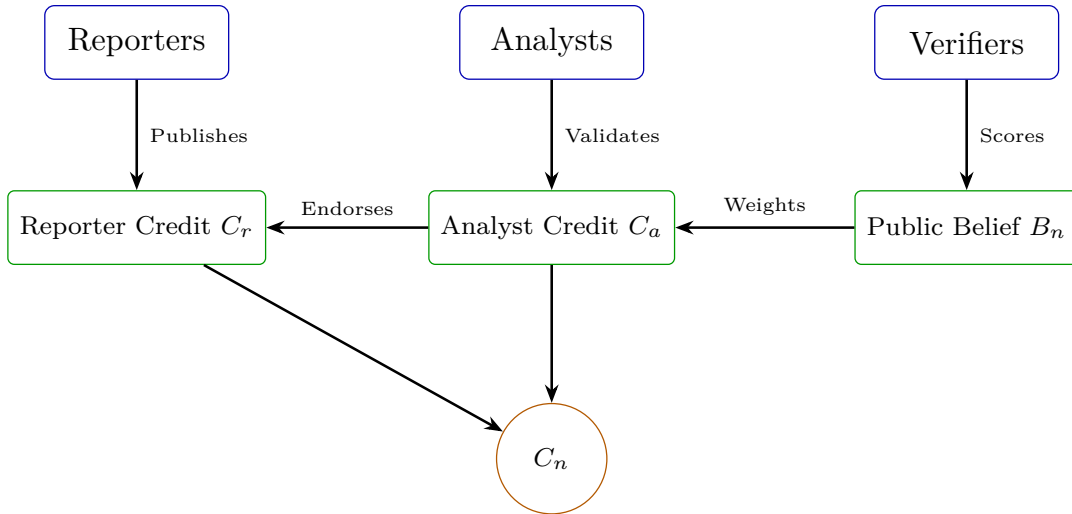


Figure 8: SANUB Reputation Flow: Relationship between roles and calculated scores

8.3 Zero-Knowledge Proof Circuits

The Privacy Pallet (146) implements zk-SNARK circuits for:

- **Humanity Proof:** Proves membership in a verified human registry without revealing identity.
- **Credential Verification:** Proves possession of credentials (e.g., press pass, organizational membership) without disclosure.
- **Age/Jurisdiction Proof:** Proves attributes meet requirements without revealing specific values.

9 Security Considerations

9.1 Threat Model

The GONO Protocol (100) is designed to resist:

- **State-Level Censorship:** Distributed validation and Polkadot shared security prevent single-point censorship.
- **51% Attacks:** Pooled security from Polkadot Relay Chain raises attack costs significantly.
- **Content Manipulation:** Cryptographic hashing ensures tamper-evidence for all indexed content.
- **Identity Exposure:** zk-SNARK proofs protect creator identity in high-stakes scenarios.
- **Sybil Attacks:** Humanity verification through Privacy Pallet (146) integration with BrightID or similar services.

9.2 Slashing Conditions

Validators are subject to slashing for:

1. Double-signing or equivocation in consensus;
2. Censorship of valid transactions;
3. Submitting fraudulent verification attestations;
4. Extended downtime beyond SLA thresholds.

10 Exemplary Use Cases

10.1 Decentralized Journalism Platform

The Gono Moncho News dApp exemplifies the protocol's primary use case:

1. A journalist captures content using the Capture SDK, which generates hashes and C2PA metadata.
2. For sensitive stories, the journalist generates a zk-SNARK proof of their press credentials without revealing identity.
3. The content is optionally stored permanently on Arweave via the Store Pallet (142).
4. Community verifiers and AI oracles assess the content through the Verify Pallet (144).
5. The final state is certified on-chain via ERC-7053.
6. Readers verify authenticity using the Gono Explorer.

10.2 AI Data Marketplace

The x402 Micropayment Pallet (148) enables automated AI data commerce:

1. An AI agent discovers a data resource protected by x402 paywall.
2. The agent receives payment requirements in the HTTP 402 response.
3. The agent submits micropayment and receives data access.
4. The transaction is recorded on-chain for provenance tracking.

10.3 Digital Asset Provenance

For digital art and collectibles:

1. The creator captures the original work with cryptographic hashing.

2. The work is stored permanently on Arweave.
3. Ownership transfers are recorded via ERC-7053 indexing.
4. Authenticity can be verified at any time through the Gono Explorer.

11 Conclusion

The GONO Protocol provides a comprehensive, modular blockchain infrastructure for verifiable media and autonomous commerce. Through its layered architecture, pluggable pallets, and integration with established standards (ERC-7053, x402, C2PA, zk-SNARKs, SANUB), the protocol offers a flexible foundation for diverse applications ranging from high-security journalism to AI data marketplaces.

The protocol's design enables selective adoption of functionality while maintaining consistent security guarantees through Polkadot shared security. Key innovations include:

- **SANUB Reputation Framework:** A mathematically rigorous system for content credibility calculation through community participation, with formalized roles for Reporters, Analysts, and Verifiers.
- **Multi-Provider Storage:** Redundant permanent storage across Arweave, Filecoin, and StorJ with Storage SLA guarantees and dynamic Top-up endowment mechanisms.
- **Gono Sovereign Governance:** A three-tier hybrid governance model combining token-weighted economic governance, reputation-weighted integrity governance, and AI-native autonomous commerce governance.
- **Decentralized Dispute Resolution:** Integration with Kleros arbitration for contested content verdicts, ensuring truth determination through game-theoretic incentives.

The GONO economic model aligns incentives across network participants, ensuring sustainable operation and governance. Through the Journalistic Integrity Council and conviction voting mechanisms, the protocol establishes a foundation for decentralized truth-seeking that resists both centralized manipulation and plutocratic capture.

12 Glossary of Terms

C2PA

Coalition for Content Provenance and Authenticity. An industry standard for embedding provenance metadata in digital content.

dAPI

Decentralized Application Programming Interface. An API provided through decentralized infrastructure.

ERC-7053

An Ethereum Request for Comments standard for on-chain content indexing and metadata management.

M2M

Machine-to-Machine. Communications or transactions between automated systems without human intervention.

Pallet

A modular component in the Substrate blockchain framework that encapsulates specific functionality.

Parachain

A blockchain that runs in parallel to the Polkadot Relay Chain, benefiting from shared

security.

SANUB

Sharing and Analyzing News Using Blockchain. A framework for content credibility calculation and fake news detection through community participation, as defined by Balouchestani et al. [1].

Substrate

A blockchain development framework created by Parity Technologies for building Polkadot-compatible chains.

x402

A protocol standard reviving HTTP 402 (Payment Required) for machine-to-machine micropayments.

zk-SNARK

Zero-Knowledge Succinct Non-interactive Argument of Knowledge. A cryptographic proof system enabling verification without revealing underlying data.

Conviction Voting

A governance mechanism where voters can multiply their voting power by voluntarily locking tokens for longer durations, weighting commitment alongside stake.

Kleros

A decentralized arbitration service using crowdsourced jurors and game-theoretic incentives for dispute resolution.

A Reference Numerals

Numeral	Component
100	Gono Protocol
110	Shared Security Layer
120	Gono Execution Rail
140	Modular Service Pallets
142	Store Pallet
144	Verify Pallet
146	Privacy Pallet
148	x402 Micropayment Pallet
160	Application Layer
210	Capture Stage
220	Store Stage
230	Verify Stage
240	Certify Stage
250	Check Stage
310	AI Agent
320	x402 Gateway
330	Service Provider
340	Gono Chain
400	GONO Token
410	Network Gas Function
420	Storage Endowment Function
430	x402 Settlement Function
440	Validator Staking Function
450	Governance Function

Numeral	Component
460	pallet-democracy
461	pallet-reputation
462	pallet-integrity-collective
463	pallet-x402-governance
464	pallet-kleros-bridge

References

- [1] A. Balouchestani, M. Mahdavi, Y. Hallaj, and D. Javdani, “SANUB: A new method for Sharing and Analyzing News Using Blockchain,” in *16th International ISC Conference on Information Security and Cryptology (ISCISC)*, Ferdowsi University of Mashhad, Iran, Aug. 2019, pp. 139–143.
- [2] Numbers Protocol, “ERC-7053: Interoperable Digital Media Indexing,” Ethereum Improvement Proposals, 2023. <https://eips.ethereum.org/EIPS/eip-7053>
- [3] Coalition for Content Provenance and Authenticity, “C2PA Specification,” Linux Foundation, 2024. <https://c2pa.org/specifications/>
- [4] x402 Foundation, “x402: Open Standard for HTTP-Native Payments,” 2024. <https://x402.org>
- [5] F. Ast and C. Lesaege, “Kleros: Short Paper v1.0.7,” Kleros, 2019. <https://kleros.io/whitepaper.pdf>
- [6] Polkadot Network, “OpenGov: Polkadot’s Decentralized Governance,” Polkadot Wiki, 2024. <https://wiki.polkadot.network/docs/learn-polkadot-opengov>